

Juan Manuel Ahumada Vázquez

AWS Solutions Architect / Senior DevOps Engineer

Open to remote — US · LATAM · EU time zones

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PROFILE 01

Computer Systems Engineer with over **10 years** of IT experience, specialized in designing, deploying and operating AWS cloud architectures under the **Well-Architected Framework**. I am responsible for the infrastructure and full life-cycle of a fleet of enterprise applications running 24/7 for a cross-border trucking company (150 employees, 370 tractor-trucks, 500 trailers), and for the AWS infrastructure of a portfolio of consulting clients. My practice spans cloud architecture, DevOps automation, site reliability engineering, security and cost optimization, with **8+ years of TypeScript / Node.js** and **6+ years of Terraform**, and extends to **production LLM systems**.

TECHNICAL STACK 02

CLOUD & INFRA	AWS · ECS / Fargate · RDS · S3 · Lambda · CloudFront · Route 53 · IAM · Secrets Manager · Organizations · Terraform (6+ yrs) · CloudFormation · CDK
CI/CD & OBSERVABILITY	Docker · GitHub Actions · multi-repo workflows · blue-green · self-hosted runners · CloudWatch · Datadog · Grafana · Sentry
LANGUAGES & FRAMEWORKS	TypeScript (8+ yrs) · Node.js (8+ yrs) · Next.js / React · Python · Go · Bash · Java / Spring Boot
DATA	PostgreSQL · pgvector · MySQL · Drizzle ORM · DMS / CDC · schema & indexing
AI & LLM	RAG · tool-calling agents · embeddings & reranking · LLM evals / LLM-as-judge · Bedrock · Claude Code

EXPERIENCE 03

RAJO Transportes · Senior DevOps Engineer & AWS Solutions Architect

Jan 2025 — Present · Nuevo Laredo, MX

Responsible for the AWS cloud architecture of a fleet of **6+ enterprise applications** in 24/7 operation. End-to-end lifecycle: architecture, IaC, CI/CD, security, observability, cost and production reliability.

KEY ACHIEVEMENTS

- **COST** Own AWS environments with monthly budgets between `$350` and `$3,400 USD`; brought the main account from `$614` to `~$400/month` (`−35%`) through right-sizing, compute optimization and storage lifecycle policies.
- **PERFORMANCE** Cut **cost per request by 71%** and median latency from `2.0 s` to `0.56 s`; rebuilt two unusable list endpoints from `19.5 s → 6 ms` and `31.8 s → 13 ms`, and brought first data on a slow mobile connection from `~21 s` to `5.0 s`.
- **RELIABILITY** Multi-repo CI/CD across 5 systems with **zero-downtime deployments and automated rollback**; **offline-first synchronization** for mission-critical field operations under intermittent connectivity; `49` CloudWatch alarms with category-based triage runbooks.
- **SECURITY** Closed an unauthenticated endpoint exposing `47,651` records (`~4,000` payroll settlements with names and salaries) and an authorization flaw across `17+` routes that let a driver act as another; remediated a PII exposure in the assistant. Least-privilege IAM, end-to-end encryption, and migration from direct prod-DB access to a **staging-readonly** model.
- **TYPESCRIPT / NODE.JS** Build **TMS RJ** with the RJ team: an in-house transportation platform (Next.js, React, Drizzle ORM, PostgreSQL) with `3` companion apps in 24/7 production and a **multi-tenant carrier portal** served from the same deployment via host-based routing.

- **AI / LLM PLATFORM** Built the TMS's **production RAG assistant**: pgvector with OpenAI embeddings, reranking, a tool-calling agent with text-to-SQL, a **groundedness gate** blocking unsupported answers, and retrieval scoped to the asker's own permissions. Re-engineering frequent questions into deterministic metrics raised accuracy `5 points` and cut misrouting `79%`, with an **eval harness using LLM-as-judge** running in CI against production.

Wake Code · Solutions Architect & DevOps Lead

2018 — Present

AWS infrastructure and SaaS products for a portfolio of **5+ clients across multiple industries** (consumer mobile SaaS, hospitality POS multi-branch, e-commerce, logistics, real estate, fitness). End-to-end ownership: architecture, IaC, CI/CD, database performance, incident response, observability and cost, with quarterly reporting to business owners.

KEY ACHIEVEMENTS

- **COST AT SCALE** Took a multi-account AWS Organizations estate (consumer mobile SaaS, `~3,000` users, two regions) from `$3,492` to `~$850/month` — a `76%` reduction held for three consecutive quarters, with `~$982/month` in standing savings.
- **EOL MIGRATION** Removed a `~$33,000/year` **end-of-support charge** by upgrading `18` RDS MySQL instances to 8.4 in four waves ordered by criticality — snapshot per wave, planned `3–9 minute` windows and post-migration data validation — reaching `$0` within the same quarter without dropping below `99.95%` availability.
- **ROOT CAUSE** Traced `~66,000` daily errors dating back to 2022 to a recursive S3 access-logging loop that had accumulated `463M` objects, and proved it with a controlled single-bucket experiment before proposing the fix.
- **ECS MIGRATION** Migrated 3 production microservices (Java/Spring Boot) and 4 batch processes from legacy VPS to ECS Fargate with **blue-green zero-downtime** deployments; `~30%` (`~$794/mo`) reduction through compute right-sizing, RDS IOPS tuning and lifecycle policies.
- **DB PERFORMANCE** Resolved a critical production deadlock on a `4.2M+ row` table with a composite index (full scan `4.2M → 34 rows`); `85%` reduction in connection leaks (`137/h → 21/h`) via pool tuning across 55 services; **AWS DMS** pipeline with `22 tables` in daily CDC; RCA of a `7h` connection-pool exhaustion outage.
- **LEGACY TO STATIC** Converted a public university campus's PHP site to a static AWS deployment, with fidelity verified route-by-route across `241` internal routes.

TecNM Campus Nuevo Laredo · Faculty Member

2019 — Present · Nuevo Laredo, MX

- Teach advanced programming at the undergraduate level in Computer Systems Engineering, and mentor students on professional residency and software development projects.

EDUCATION 04

2013 — 2018

TecNM Campus Nuevo Laredo

B.Sc. Computer Systems Engineering

Honorable mention for professional residency project; represented the National Technological Institute of Mexico — Nuevo Laredo campus — at a national technological innovation competition.

CERTIFICATIONS 05

AWS Certified Solutions Architect — Associate `SAA-C03 · in preparation`

Exam targeted 2027.

LANGUAGES 06

Spanish	Native
English	Professional